

Modeling the Local Economic Impact of Datacenters at the US County Level

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Research Question

How does the deployment of a new data center impact the local economic conditions such as employment, prices, and business development at a county level in three years of construction?

Motivation

While the first data center was opened in 1946,¹ it was not until the 2000s that they started to become common, and only a handful of years since an explosion in both construction and scale of new data centers due to the widespread adoption of LLMs.¹ This has led to a dramatic increase in the amount of energy and water used, space needed for these developments, and potential economic impact.² Most of the public discourse and recent research on data centers revolve around the impacts on energy and water. However, the economic impact on the communities surrounding these data centers has received relatively little attention to date.

Broad discussions are happening around how data centers may impact the national or global economy, specifically surrounding data centers' contribution to the growth of AI. Unfortunately, these broader impacts often take longer to permeate through the economy, and thus may prove difficult to study until more time or data becomes available. Additionally, it is widely acknowledged that the LLMs that data centers support have had a major impact on the technology people are using. Instead, we'd like to take the more local approach of studying how the rapid construction of these massive computing campuses can have immediate effects on a community.

As mentioned above, the energy and water requirements of data centers have already garnered significant public and research attention, and the environmental impacts on communities are negative.^{3,4} In this project, we intend to expand the discourse to include not only environmental, but also economic factors such as employment and business growth. This has the advantage of broadening the scope of how people think about data centers to draw a more holistic picture of what their overall impacts may be at a community level. We believe this approach could help drive further political discussion around what should be done about data centers, or may motivate people to think about potential alternatives to their construction.

Data

Inspired by a review of existing local economic impact analyses, we decided to only include data sets that have a granularity at the US county level. Compiling county-level

datasets, we found that the most relevant data covered the years between 2001 and 2023, which naturally formed our second selection criterion.

To establish the location of data centers in the US we plan to utilize a free-trial period to access the Cleanview data center locations API ([link](#)) which provides longitude and latitude coordinates. These coordinates include currently existing and planned data centers as well as dates of construction. If construction dates are missing for any locations, we plan to manually find this using Google Maps' timeline functionality.

Our economic features will include unemployment and median household income by county from the USDA ([link](#)), Residential construction permit counts ([link](#))([link](#)), population estimates ([link](#)), GDP ([link](#)), employment/business patterns by industry ([link](#)), and personal income([link](#)). All of these data are listed at the county level annually for our analysis timeframe. We will be downloading each of these datasets as CSVs or Excel files then uploading them to a normalized Postgres database. Each observation will consist of a unique key-year combination.

Our economic data will be joined with our datacenter locations using PostGIS to assign a county to the datacenter coordinates.

Methodology Requirements

Statistical Thinking (DATA 500 and 501)

Once we have loaded our data into a SQL database, we will conduct exploratory analysis by checking summary statistics for all of our features to try and spot trends and or outliers which need further attention. If we find small gaps in our time-series data we may infer those values using the average of the values that come before and after the gap. We will also be normalizing all nominal monetary values to adjust for inflation.

We plan to use correlation to check the strength of the relationships between our explanatory and predictor variables. This will act as a test before we include any features in machine learning models. In a similar way, we plan to use statistical measures like the r-squared and statistical significance of model coefficients to assess how confident we ought to be in our findings.

Data Visualization (DATA 502)

For this project, we aim to leverage visualizations to support both our findings and people's understanding of the issue as a whole. One of the major visualizations that will help us do this will be a map of the entire United States with data centers as points on the map, which will help people understand the scale and widespread implementation of these centers. It may also be helpful to color the points by year (or year clusters) to give a clearer picture of how much these have exploded in recent years.

Another similar visualization would be to do a choropleth map of some of the economic variables to visualize regional trends and differences. Since we will be looking at US counties, it may be helpful to visualize how economic variables can differ regionally, which will also help support our reasoning for not just doing this at a national level.

To further increase people's baseline understanding of our topic area, we will probably also want to include some line graphs of many of the economic variables that we're using (such as unemployment rate over time or inflation rate over time). This will help people see how these things trend over time, and will also allow us to show how dramatic (or insignificant) the impact has been on each of these variables individually since data centers started cropping up en masse.

The next logical step for us here will be to actually do a regression analysis and graph the results, to establish a more statistically grounded analysis of these impacts. We'll likely want to include the scatterplots of points as well as the regression lines for these.

Data Engineering (DATA 503 and 513)

The initial collection and cleaning of our data will kick off the data engineering portion of this project. We hope to organize all of our datasets into SQL databases for easy querying and EDA, so we will utilize PostgreSQL to put our data into staging tables and then normalize and clean those tables. We will then extract those tables into CSV as needed for use in Python for our data analysis and machine learning implementations.

Additionally, we may end up having to utilize some other data engineering techniques for our data center dataset. As mentioned in the data section, we are hoping to use the Cleanview data center dataset as it contains year constructed for time-series analysis. However, we are only allotted a free trial of 7 days, and are unsure yet of whether we will be able to get the appropriate data within this time frame. If not, there is an API available that we could potentially explore, or we may have to find a more general dataset (which doesn't include construction date) and manually add the construction dates using Google Earth.

Machine Learning (DATA 505 and 515)

Linear regression is the standard modeling technique in economics literature so we plan to use them to create a baseline condition against which we can compare models using other machine learning techniques. We are interested in understanding how the presence of a datacenter might impact the economic conditions of a county so we will explore multiple linear models predicting values such as median income, rates of business creation, etc. During the model exploration phase we will engineer features including but not limited to using PCA. From those models we will select the strongest based on accuracy, precision, and our understanding of economic principals.

Then we will create competing models using neural networks to predict the same outcome as our linear model(s) and compare the performance differences. We will compare model complexity and generalizability in addition to accuracy when determining the better approach.

Data Ethics (DATA 504)

Overall there are several ethical considerations to take into account when looking at data centers. As we've talked about previously, data centers have frequently been the target of

ethical debates, as they have a known negative environmental impact. We will want to consider this when forming our conclusion, as even if we don't find negative economic impacts, this does not mean that data centers are without harm, and we must be conscious of that.

In addition to the topic as a whole, many of the variables that we'll be using for this project require ethical consideration during discussion and analysis. For example, we hope to look at demographic data, but we need to be cognizant of when it is appropriate or not to actually use this. For example, training models on race, sex, or other sensitive demographics can lead to biased models that perpetuate negative stereotypes, so we may want to avoid using these factors in any machine learning model unless we are certain that they have a real and important impact.

Economic data inherently has certain ethical factors that are important to acknowledge. While the economy is a broad concept, its implications can be very personal. When we talk about high unemployment, we are also saying that there are people who are potentially struggling to make ends meet because they cannot find a job. Similarly, high inflation means that people may not be able to buy the things that are necessary for them to live happy and healthy lives. It's important for us to consider this when discussing the implications of these findings, as it can have very real impacts on the people living in the communities that data centers are built in. On a similar note, we will want to make sure that our terminology is appropriate and considerate of this reality (e.g. when discussing people below the poverty line we'll want to make sure to say something such as "people experiencing poverty" instead of other, more dehumanizing phrases).

Sources

1. <https://www.datacenterknowledge.com/data-center-construction/8-milestones-in-data-center-history-1946-to-today>
2. <https://www.nprillinois.org/2025-07-21/data-centers-water-electricity-growing-usage>
3. <https://www.eesi.org/articles/view/data-centers-and-water-consumption>
4. <https://www.nber.org/papers/w35100>